

Technical Report: Housing Habitability Prediction Model

1. Executive Summary

This document details the development and evaluation of a predictive machine learning model designed to estimate the habitability scores of residential properties. Utilizing a dataset of 35,549 property records, an end-to-end data processing and regression pipeline was engineered in Python. The system leverages an ensemble Random Forest algorithm to map 25 distinct environmental, structural, and categorical variables to a continuous habitability index, achieving an R^2 variance explanation of over 83%.

2. Data Ingestion and Preprocessing

The initial dataset comprised 15 variables per property, containing both continuous and categorical data types, alongside a significant volume of missing entries. To prepare the data for algorithmic ingestion, a systematic preprocessing pipeline was executed:

- **Missing Value Imputation:** Continuous features, specifically `Number_of_Windows` and `Frequency_of_Powercuts`, were imputed using the statistical median to prevent skewing from outliers. Categorical features, including `Furnishing`, `Crime_Rate`, and `Dust_and_Noise`, were imputed using the statistical mode.
- **Dimensionality Reduction:** The `Property_ID` attribute was dropped from the dataset as it served only as an arbitrary identifier and provided no predictive value.
- **Categorical Encoding:** One-hot encoding (`pd.get_dummies` with `drop_first=True`) was applied to convert text-based classifications (e.g., `Property Type`, `Water Supply`, `Crime Rate`) into binary boolean arrays. This expanded the feature space from the initial 14 predictors to a robust 25-dimensional feature matrix.

3. Model Architecture and Training

The predictive engine utilizes a `RandomForestRegressor`, a robust ensemble learning method that constructs multiple decision trees during training and outputs the mean prediction of the individual trees.

The preprocessed dataset was partitioned into an 80/20 train-test split, yielding 28,439 samples for training and 7,110 samples for validation. The Random Forest model was initialized with 100 estimators (`n_estimators=100`) and configured to utilize all available CPU cores (`n_jobs=-1`) to accelerate the training phase. Following model convergence, the engineered feature columns, training subsets, and the fully trained estimator were serialized using the `joblib` library to facilitate seamless downstream deployment and reproducible inference.

4. Evaluation and Performance Metrics

The serialized model and holdout test datasets were reloaded to evaluate predictive performance against unseen data. The model's accuracy was quantified using two primary statistical metrics:

- **Coefficient of Determination (R^2):** The model achieved an R^2 score of 0.8319. This indicates that the algorithm successfully captures and explains 83.2% of the variance in housing habitability scores based on the provided structural and environmental features.
- **Mean Absolute Error (MAE):** The model recorded an MAE of 4.5010. On average, the algorithm's predictions deviate from the true empirical habitability score by only 4.50 points, demonstrating high precision for real-world real estate estimation.

5. Conclusion

The implementation of the Random Forest regression pipeline successfully established a high-accuracy, data-driven approach to evaluating residential property habitability. The robust handling of missing values and the utilization of multi-core ensemble processing ensure that the model is both computationally efficient and highly scalable for larger real estate datasets.